



Sai Bharadhwaj Matha

Research Engineer in AI, Computer Vision and Embedded Systems

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Professional Summary

Research Engineer in AI, Computer Vision, and Embedded Systems with 3+ years of cross-disciplinary experience in embedded systems, 3D computer vision, and robotics, including contributions to real-world UAV systems and advanced deep learning models. Skilled in developing semantic and panoptic occupancy prediction pipelines, SLAM-based UAV navigation, and multi-modal perception frameworks. Seeking to leverage expertise in 3D computer vision, AI for autonomous systems, and agentic frameworks to advance innovation in robotics and intelligent perception.

Education

2022 – 2025 Cham, Germany	Master of Engineering in Artificial Intelligence for Smart Sensors and Actuators <i>Technische Hochschule Deggendorf.</i> GPA: 1.4 / 5.0 (awarded as the best final grade of the master's program) Thesis: Real-world Semantic Occupancy Prediction for Advanced Air Mobility.
2016 – 2020 Rourkela, India	Bachelor of Technology in Electrical Engineering <i>National Institute of Technology Rourkela.</i> CGPA: 8.72 / 10.0 (corresponds to 1.7 on German grading scale) Thesis: Wireless Power Transfer and Its Application in Solar Power Harvesting.
2014 – 2016 Visakhapatnam, India	Higher Secondary Education <i>Board of Intermediate Education Andhra Pradesh.</i> Percentage: 98.4

Professional Experience

11.2025 – Present Ingolstadt	Research Associate <i>Full-time, Technische Hochschule Ingolstadt</i> - Developed a marker-based landing function using OCTAS and AirSim simulations. - Built a ROS2 package for a topology-aware emergency landing system with 95% success rate using SegFormer (70.22% mIoU) and metric depth at 12 FPS. - Engineered custom models and a novel low-altitude dataset for dynamic object detection.
02.2024 – 09.2025 Ingolstadt, Germany	Research Assistant <i>Part-time, Fraunhofer IVI</i> - Designed a novel 3D semantic and panoptic scene completion data pipeline for UAS. - Engineered a 50K+ sample dataset with RGB, thermal, depth, and 3D occupancy GT. - Trained SOTA SSC models (DISC, CGFormer...) to develop a novel 3D architecture. - Cut external annotation costs by 97% via image subset optimization.
09.2020 – 09.2022 Mumbai, India	Embedded Systems Engineer <i>Full-time, Ideaforge Technology Private Limited.</i> - Headed the development of the propulsion system, ensuring reliable performance. - Engineered an FOC-based ESC for BLDC motors and a Li-ion battery pack charger. - Developed embedded HW/SW for GPS-denied UAV navigation via ORB-SLAM3.

Key Technical Projects

02.2026 – Present	Real-time Visual SLAM Pipeline for UAV swarms - Implementation of a collaborative visual SLAM pipeline (PX4 SIL, ORB-SLAM3, OpenCV, ROS 2) for UAS in a simulation environment (Gazebo Classic/AirSim). - Working on a novel multi-agent constellation extension for collaborative SLAM.
11.2025 – 03.2026	MonoSpatial: Agent-Based 3D Spatial Interaction in Monocular RGB Images - Developed an agentic reasoning pipeline to orchestrate vision models for 3D spatial queries. - Leveraged a 17B MoE LLM via MCP to orchestrate multi-step monocular-to-3D geometric inference pipelines. Utilizes YOLO-world with Depth Anything v3 and FastMCP servers.
03.2025 – 06.2025	Multi-Modal 3D Object Detection in Adverse Weather Conditions - Designed and trained a deep autoencoder for 2D feature extraction in adverse weather. - Implemented early-fusion for multi-modal synthetic data generated in CARLA.

3D Semantic and Panoptic Occupancy Prediction for Advanced Air Mobility

- A novel benchmark SSC and PSC dataset for UAS, ran ablations on SOTA models, and a novel geometry-aware model architecture for aerial scenes.
- Designed a data-generation pipeline that integrates 3D reconstruction, pose estimation, semantic fusion, mesh generation, voxelization, extrusion and voxel densification.

Technical Skills

Languages: Python, Rust, C++, SQL, Bash/Linux.

AI & Vision: PyTorch, ML/DL, 2D/3D Computer Vision, GenAI, Agentic AI (MCP).

Robotics: ROS/ROS2, STM32, RTOS, UAV Avionics, Power Electronics, Sensor Fusion, edge-AI.

Tools/Cloud: Docker, Kubernetes, Git, ONNX, TensorRT, AWS, Microsoft Ecosystem.

Publications

OccuFly: A 3D Vision Benchmark for Semantic Scene Completion from the Aerial Perspective

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

SegFly: A 2D-3D-2D Paradigm for Aerial RGB-Thermal Semantic Segmentation at Scale

European Conference on Computer Vision (ECCV), 2026

SafeLand: Safe Autonomous Landing in Unknown Environments with Bayesian Semantic Mapping

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026

Courses & Certificates

Big Data

Issued by Coursera

Quantum Computing

Elective from THD

MLOps (AWS): Deploying AI & ML Models

Issued by edX

AI Agents and MCP

Issued by Hugging Face

Languages

English — Full Professional Proficiency

Telugu — Native/Bilingual

German — Conversational

Hindi — Native/Bilingual

Leadership and Volunteering

2019 – 2020

Rourkela, India

VS Hall of Residence

Student elected representative

2017 – 2020

Rourkela, India

Plugged_IN

Vice President

Awards

2025

Award for Best Student, M.S.S. Master's Program

Förderverein of Campus Cham, Technische Hochschule Deggendorf.

Interests

• Cooking and blogging

• eFootball and gaming

References

Prof. Dr. Dmitrii Dobriborsci, *Professor*, Technische Hochschule Deggendorf

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Henri Meess, *M.Sc., VP of AI & Autonomy*, SWARM Biotactics

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Declaration

I affirm that all information provided is true and accurate to the best of my knowledge.

M. Sai Bhaskarhuja.

Ingolstadt, 30.04.2026